

UST-KAMCHATSKY, Russian Federation

WMO: 324080

Lat: 56.2369N

Lon: 162.6919E

Elev: 99

StdP: 14.64

Time Zone: 12.00 (E12)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.7	-14.6	-24.8	1.4	-19.3	-20.0	1.8	-14.0	36.5	28.9	32.5	28.7	5.9	350	0.812

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.5	69.2	59.6	65.6	57.6	62.5	55.9	60.9	67.3	58.7	63.7	56.8	61.2	8.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.9	71.8	63.3	56.2	67.4	61.0	54.6	63.6	59.0	27.1	68.1	25.6	64.0	24.4	61.3	69.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.5	25.0	21.4	DB	-25.7	76.4	5.0	3.5	-29.3	78.9	-32.3	81.0	-35.1	83.0	-38.7	85.5	(4)
(5)			WB	-26.9	64.8	5.2	2.3	-30.6	66.4	-33.6	67.7	-36.5	69.0	-40.3	70.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.3	10.8	11.8	19.1	26.9	35.6	46.4	53.8	54.8	48.8	37.8	26.0	14.9		
	DBStd	17.78	14.57	13.14	10.86	6.62	4.01	5.20	4.74	3.99	4.74	5.89	8.41	11.92		
	HDD50	6794	1215	1070	956	693	446	133	13	6	77	378	719	1088		
	HDD65	11919	1680	1490	1421	1143	911	557	348	317	487	843	1169	1553		
	CDD50	350	0	0	0	0	0	26	130	154	40	0	0	0		
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH74	17	0	0	0	0	0	2	12	3	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	9.3	10.1	11.1	10.2	10.0	8.6	9.5	8.9	8.3	8.1	8.5	8.7	9.3		
(15) Precipitation	PrecAvg	27.6	3.1	2.4	2.2	1.4	1.1	1.4	1.6	2.5	2.3	2.9	2.9	3.4		
	PrecMax	34.2	5.9	5.1	3.6	2.6	2.6	3.1	3.0	4.7	7.2	5.2	5.2	6.3		
	PrecMin	18.1	0.5	0.2	0.5	0.4	0.2	0.0	0.0	0.8	0.5	1.3	0.9	1.7		
	PrecStd	3.3	1.5	1.3	0.9	0.6	0.6	0.8	0.8	1.1	1.5	1.2	1.2	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.8	35.7	36.7	40.9	54.0	71.1	75.5	72.8	65.0	53.0	44.9	39.6		
		MCWB	35.0	33.8	34.2	35.3	44.8	56.2	62.3	62.6	54.9	44.7	41.8	37.8		
	2%	DB	34.1	33.4	34.9	37.8	48.6	64.7	70.0	68.6	61.5	50.6	42.2	35.8		
		MCWB	32.8	32.3	33.0	33.2	41.5	53.2	60.5	60.0	53.6	44.7	39.9	33.7		
	5%	DB	32.7	32.0	33.6	35.9	44.8	60.1	66.0	65.1	58.9	48.6	40.0	33.6		
		MCWB	31.7	31.2	32.1	32.7	39.6	51.4	58.7	58.0	52.4	44.0	37.5	32.2		
	10%	DB	31.0	30.3	32.3	34.4	41.9	56.0	62.5	62.1	56.6	46.8	37.6	31.7		
		MCWB	30.1	29.3	31.0	32.0	37.8	49.5	56.7	56.8	51.6	42.9	35.1	30.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.1	34.2	35.0	36.3	45.5	57.2	63.7	64.2	57.4	48.6	42.8	37.8		
		MCDB	36.1	34.9	36.1	38.7	52.8	68.5	72.6	70.7	61.7	50.7	44.0	39.1		
	2%	WB	33.2	32.5	33.5	34.2	41.9	54.2	61.2	61.1	55.3	46.7	40.2	34.1		
		MCDB	33.8	33.1	34.4	36.1	47.4	62.7	68.5	66.8	58.9	48.7	41.8	35.1		
	5%	WB	31.9	31.3	32.4	33.3	40.0	51.9	59.2	58.9	53.8	45.0	37.8	32.4		
		MCDB	32.5	32.0	33.3	34.9	44.2	59.1	65.0	63.3	57.3	47.5	39.6	33.4		
	10%	WB	30.0	29.3	31.2	32.5	38.3	49.7	57.3	57.1	52.4	43.2	35.5	30.5		
		MCDB	30.9	30.2	32.2	34.1	41.5	55.4	61.8	61.0	55.7	46.1	37.2	31.6		
(35) Mean Daily Temperature Range	MDBR		13.2	13.6	13.7	11.2	8.6	11.9	11.9	11.5	13.5	12.6	11.8	13.1		
	5% DB	MCDBR	8.9	9.0	8.4	11.3	14.7	21.6	19.8	18.4	16.6	14.8	11.1	9.8		
		MCWBR	8.8	8.5	7.6	8.5	9.2	12.2	11.4	10.6	9.9	10.3	10.1	9.4		
	5% WB	MCDBR	8.5	8.3	7.8	8.6	13.4	20.4	17.9	15.7	13.5	12.1	10.2	9.4		
		MCWBR	8.6	8.1	7.5	6.7	8.7	11.9	10.9	9.7	8.9	9.9	9.8	9.4		
(40) Clear-Sky Solar Irradiance	taub		0.300	0.328	0.370	0.418	0.454	0.432	0.454	0.429	0.362	0.346	0.306	0.288		
	taud		2.047	2.047	1.971	1.911	1.966	2.183	2.158	2.227	2.401	2.271	2.148	2.032		
	Ebn at Noon		175	221	242	249	248	258	248	246	251	218	175	146		
	Edh at Noon		20	30	42	52	53	43	43	38	28	24	18	16		
(44) All-Sky Solar Radiation	RadAvg		152	361	738	1160	1449	1555	1509	1199	920	518	222	112		
	RadStd		25	89	122	156	153	170	130	105	70	42	18	16		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page